

Differential Evolution

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Outline

- Differential Evolution (DE)
- Difference-vector based mutation
- Recombination operators
- Variants of DE
- Detailed working of DE with an example
- Preliminary comparison of algorithms

Differential Evolution



[Journal of Global Optimization](#)

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Differential Evolution – A Simple and Efficient Heuristic for global Optimization over Continuous Spaces

Authors

[Authors and affiliations](#)

Rainer Storn, Kenneth Price

Article

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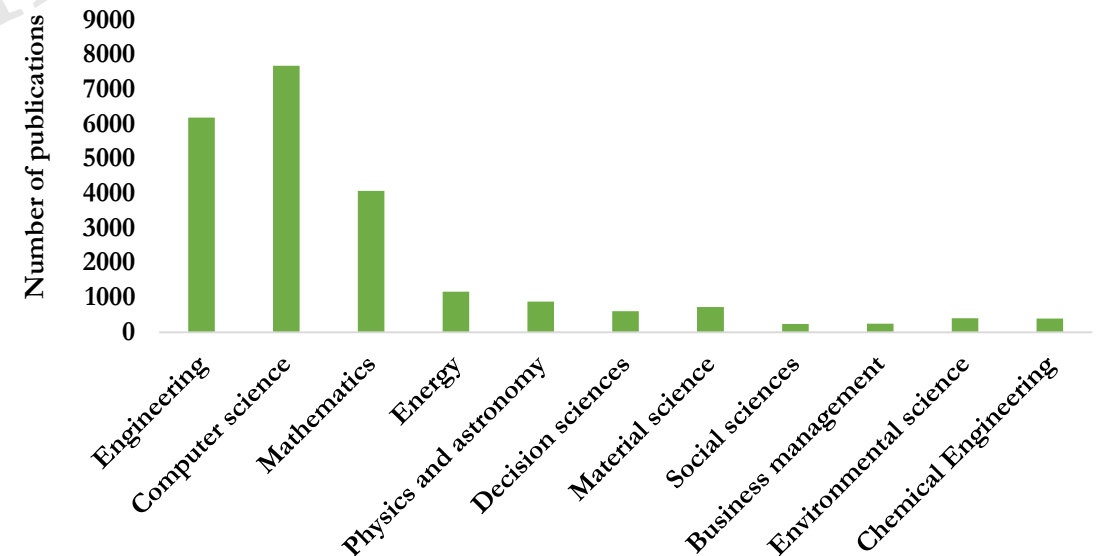
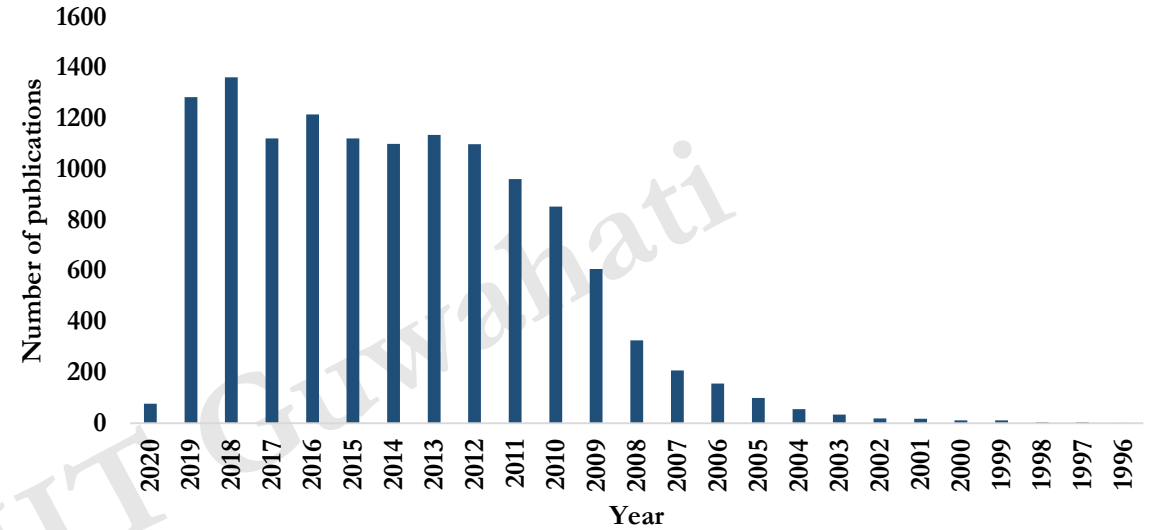
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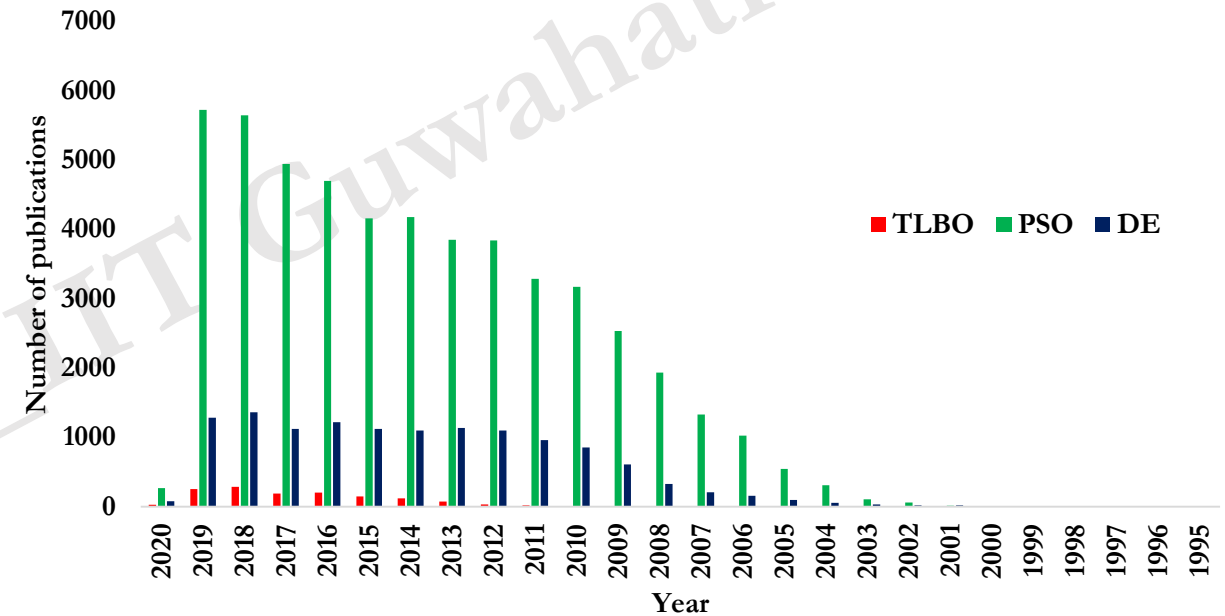
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Differential Evolution

- Stochastic population based technique.
- Each solution is known as **genome/chromosome**.
- Each chromosome undergoes mutation followed by recombination.



- Selection of better solutions is performed only after the generation of all trial vectors.
- Greedy selection is performed between target and trial vectors.

Difference-vector based mutation

➤ Donor vector (V) of a chromosome (X_i) is created as

$$V = X_{r_1} + F(X_{r_2} - X_{r_3})$$

F Scaling factor, a constant between 0 and 2

r_1, r_2, r_3 Random solutions $r_1, r_2, r_3 \in \{1, 2, 3, \dots, N_p\}$ and $r_1 \neq r_2 \neq r_3 \neq i$

➤ Target vector is not involved in mutation.

➤ Total 4 vectors are involved in the mutation of a target vector and hence $N_p \geq 4$.

Recombination: Binomial (uniform) crossover

➤ Performed to increase the diversity

➤ Creation of trial vector can be

$$u^j = \begin{cases} v^j & \text{if } r \leq p_c \text{ OR } j = \delta \\ x^j & \text{if } r > p_c \text{ AND } j \neq \delta \end{cases}$$

p_c crossover probability

δ randomly selected variable location $\delta \in \{1, 2, 3, \dots, D\}$

r random number between 0 and 1

u^j j^{th} variable of trial vector

v^j j^{th} variable of donor vector

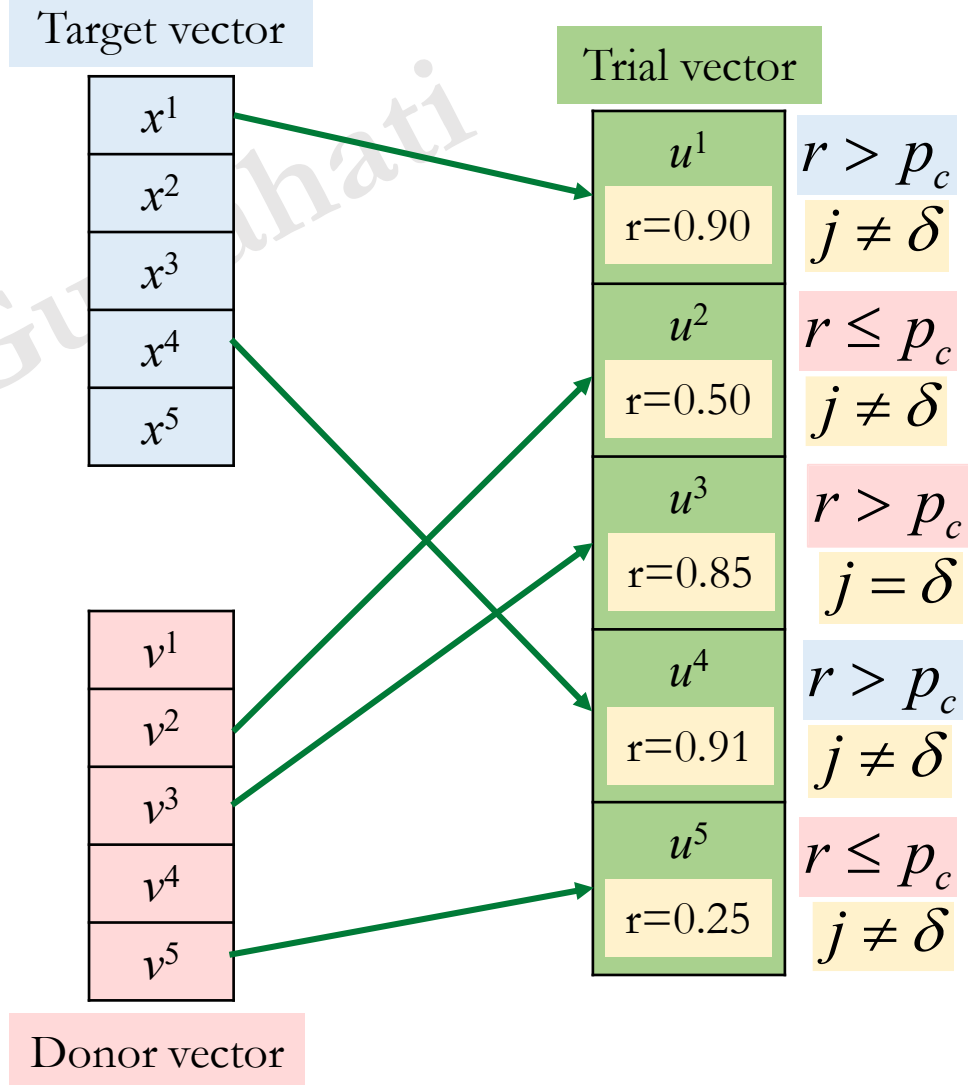
x^j j^{th} variable of target vector

➤ δ ensures that at least one variable is obtained from the donor vector

➤ Probability for crossover (p_c) is generally high

➤ High p_c results in more variables from donor

Let $D = 5$, $\delta = 3$, $p_c = 0.8$



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Let $D = 5$, $\delta = 3$, $p_c = 0.8$

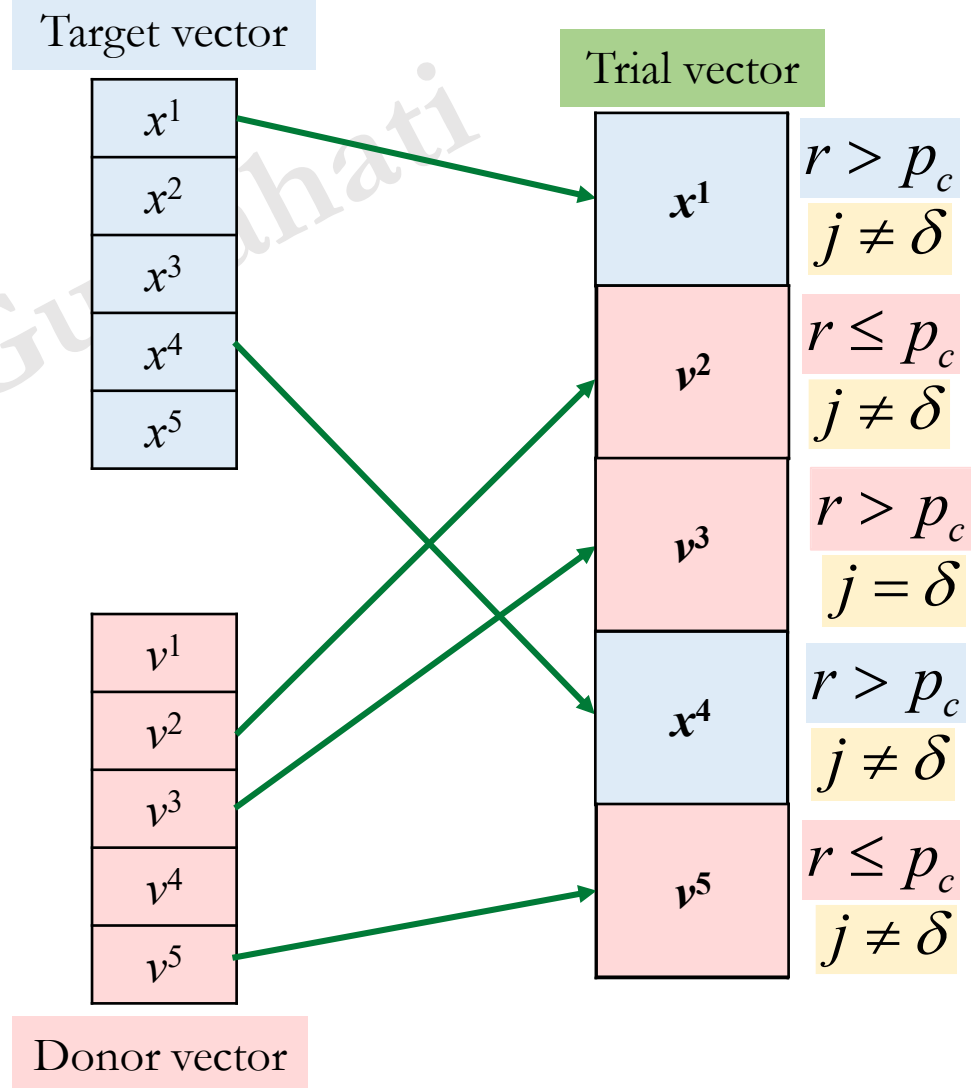
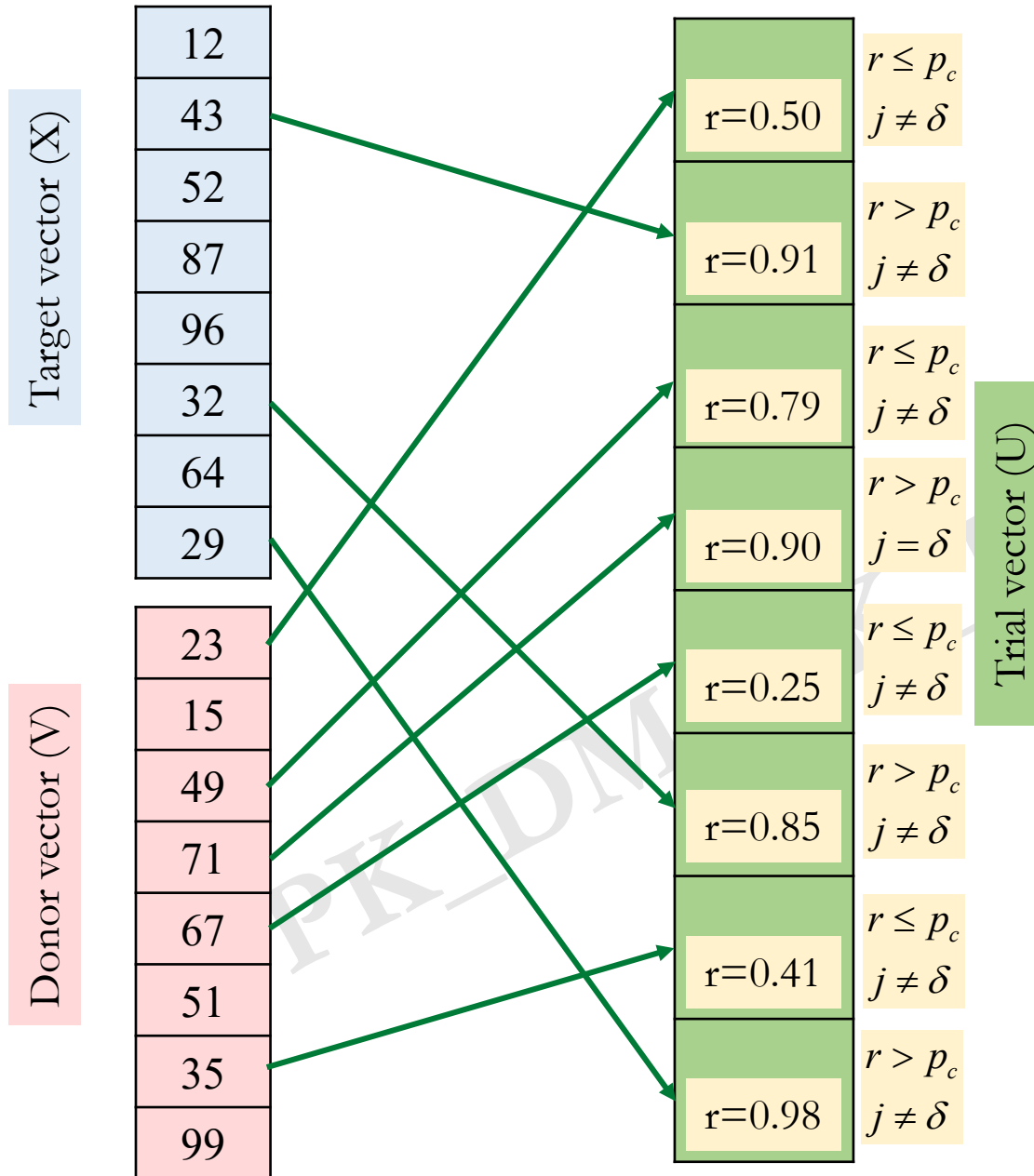


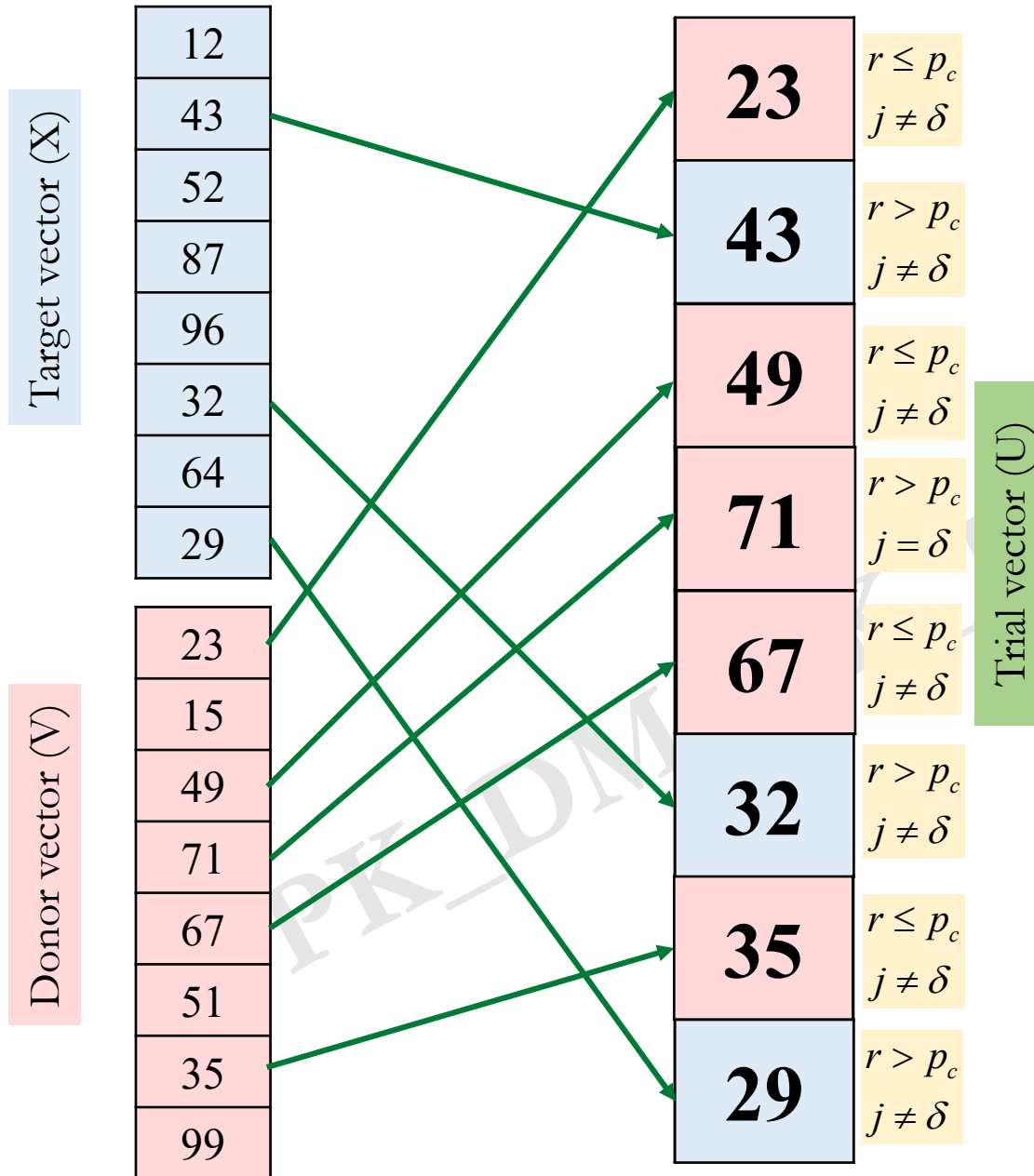
Illustration of binomial crossover



$\delta = 4$ and $p_c = 0.8$

$$u^j = \begin{cases} v^j & \text{if } r \leq p_c \text{ OR } j = \delta \\ x^j & \text{if } r > p_c \text{ AND } j \neq \delta \end{cases}$$

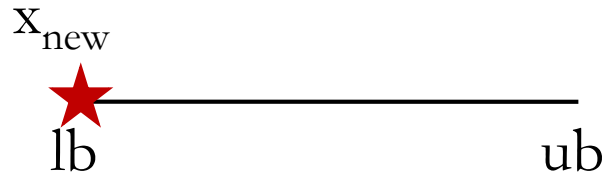
Illustration of binomial crossover



$\delta = 4$ and $p_c = 0.8$

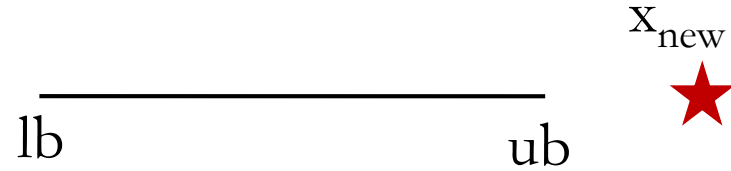
$$u^j = \begin{cases} v^j & \text{if } r \leq p_c \text{ OR } j = \delta \\ x^j & \text{if } r > p_c \text{ AND } j \neq \delta \end{cases}$$

Bounding of offspring



x_{new} is within bounds

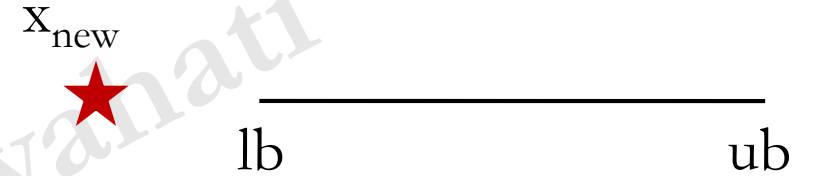
No bounding required



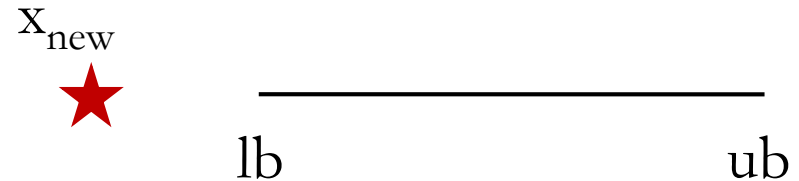
x_{new} violates the upper bound



Shift x_{new} to upper bound



x_{new} violates the lower bound



Shift x_{new} to lower bound

Selection

- Evaluate the fitness function of all offspring (f_U).
- Population is updated using greedy selection.

$$\left. \begin{array}{l} X_i = U_i \\ f_i = f_{U_i} \end{array} \right\} \text{ if } f_{U_i} < f_i$$

X and f remains the same if $f_{U_i} > f_i$

- Greedy selection is performed only after the generation of offspring by all solutions.

Pseudocode of DE

Inputs: Fitness function, lb, ub, N_p , T , F , p_c

1. Initialize a random population (P)

2. Evaluate fitness (f) of P $\leftarrow$ FE = N_p

for $t = 1$ to T

for $i = 1$ to N_p

Generate the donor vector (V_i) using **mutation**

Perform **crossover** to generate offspring (U_i)

end

for $i = 1$ to N_p

Bound U_i

Evaluate the fitness (f_{U_i}) of U_i $\leftarrow$ FE = 1

Perform **greedy selection** using f_{U_i} and f_i to update P

end

end

In one iteration, #FE = N_p

For T iterations, #FE = $N_p + N_p T$

$$V = X_{r_1} + F(X_{r_2} - X_{r_3})$$

Generation

$$u^j = \begin{cases} v^j & \text{if } r \leq p_c \text{ or } j = \delta \\ x^j & \text{if } r > p_c \text{ and } j \neq \delta \end{cases}$$

Selection

$$\left. \begin{array}{l} X_i = U_i \\ f_i = f_{U_i} \end{array} \right\} \text{ if } f_{U_i} < f_i$$

Exponential crossover

- Randomly choose an integer (n) between 1 and D .
- Copy the n^{th} variable from donor as n^{th} variable of trial vector.
- For the subsequent variables, generate a random number between 0 and 1, till $r > p_c$.
- If $r \leq p_c$, copy the variable from donor to trial vector.
- If $r > p_c$, copy the remaining variables from target to trial vector.

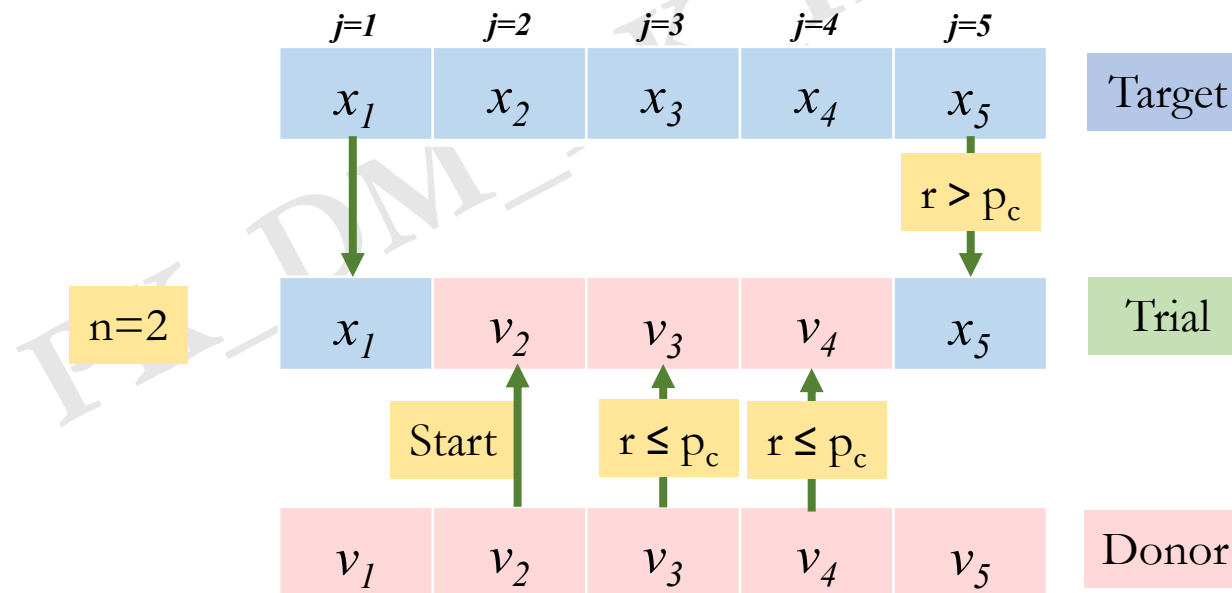
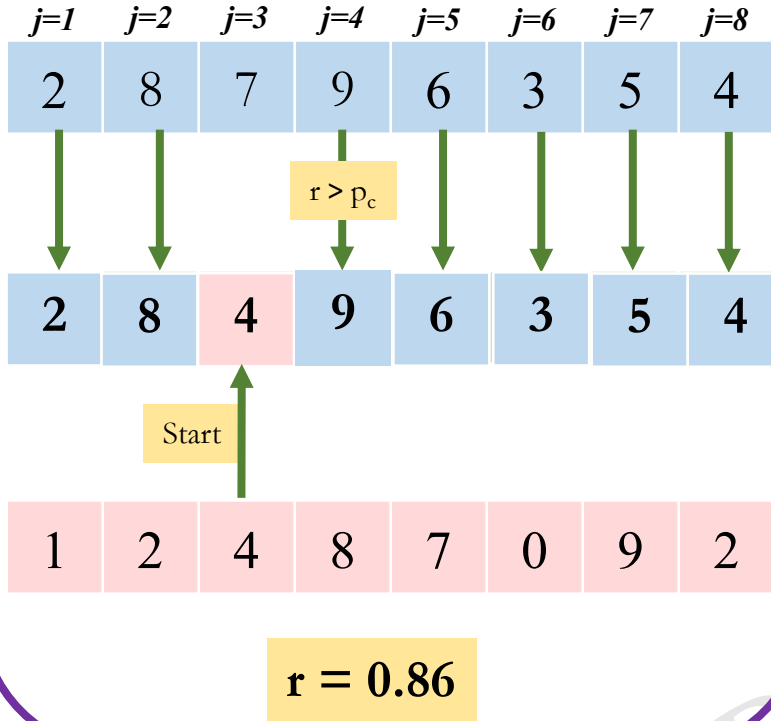
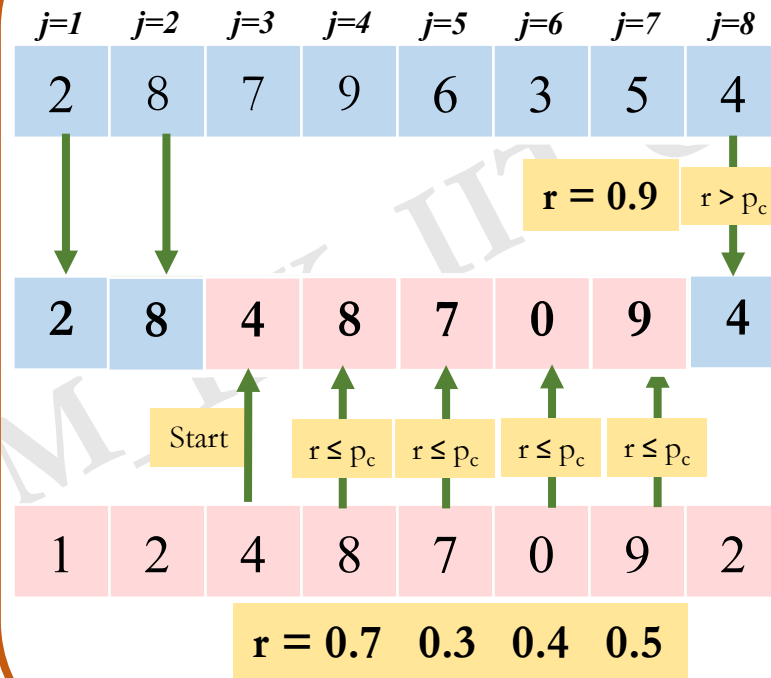


Illustration of exponential crossover

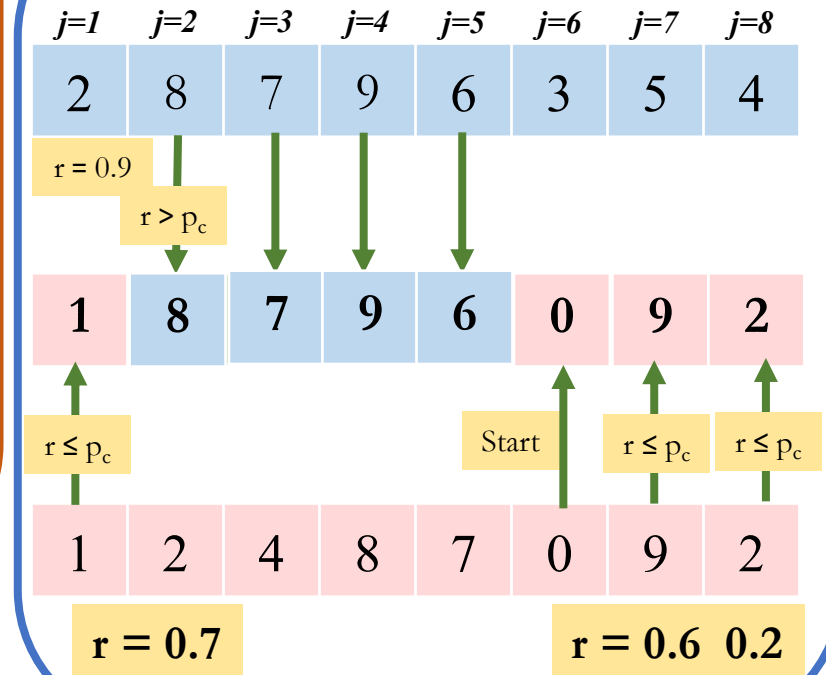
Let $n = 3, p_c = 0.8$



Let $n = 3, p_c = 0.8$



Let $n = 6, p_c = 0.8$



Mutation strategies (DE/x/y/z)

- DE: Differential Evolution
- x: Vector to be mutated
- y: number of difference vectors (random solutions) required for mutation
- z: type of crossover scheme to be used (can be either exponential or binomial crossover)

Strategy	Expression for donor vector	Minimum N_p
DE/rand/1	$V = X_{r_1} + F(X_{r_2} - X_{r_3})$	4
DE/best/1	$V = X_{best} + F(X_{r_1} - X_{r_2})$	3
DE/rand/2	$V = X_{r_1} + F(X_{r_2} - X_{r_3}) + F(X_{r_4} - X_{r_5})$	6
DE/best/2	$V = X_{best} + F(X_{r_1} - X_{r_2}) + F(X_{r_3} - X_{r_4})$	5
DE/target-to-best/1	$V = X_i + F(X_{best} - X_i) + F(X_{r_1} - X_{r_2})$	3

Working of DE: Sphere function

Consider $\min f(x) = \sum_{i=1}^4 x_i^2; \quad 0 \leq x_i \leq 10, \quad i = 1, 2, 3, 4$

$$f(x) = x_1^2 + x_2^2 + x_3^2 + x_4^2$$

Decision variables: x_1, x_2, x_3 and x_4 and dimension $D = 4$

- **Step 1:** Fix the population size, number of generation, crossover probability, scaling factor

$$N_p = 5, T = 10, p_c = 0.8, F = 0.85$$

- **Step 2:** Generate random solutions within the domain of the decision variables

$$P = \begin{bmatrix} 4 & 0 & 1 & 8 \\ 3 & 1 & 9 & 7 \\ 0 & 3 & 1 & 5 \\ 2 & 1 & 4 & 9 \\ 1 & 2 & 8 & 3 \end{bmatrix} \quad f = \begin{bmatrix} 81 \\ 140 \\ 35 \\ 102 \\ 78 \end{bmatrix}$$

Working of DE: first solution

- Step 3: Generate 4 random integers between 1 and N_p

$$\text{Let } r_1 = 4 \quad r_2 = 2 \quad r_3 = 3 \quad \delta = 1$$

- Step 4: Determine donor vector (Mutation)

$$\begin{aligned} V_1 &= X_4 + F(X_2 - X_3) \\ &= [2 \ 1 \ 4 \ 9] + 0.85 \times ([3 \ 1 \ 9 \ 7] - [0 \ 3 \ 1 \ 5]) \\ &= [2 \ 1 \ 4 \ 9] + [2.55 \ -1.7 \ 6.8 \ 1.7] \\ &= [4.55 \ -0.7 \ 10.8 \ 10.7] \end{aligned}$$

- Step 5: Generate D random numbers

$$\text{Let } r = [0.3 \ 0.9 \ 0.2 \ 0.6]$$

$$p_c = 0.8, F = 0.85$$

$$P = \begin{bmatrix} 4 & 0 & 1 & 8 \\ 3 & 1 & 9 & 7 \\ 0 & 3 & 1 & 5 \\ 2 & 1 & 4 & 9 \\ 1 & 2 & 8 & 3 \end{bmatrix} \quad f = \begin{bmatrix} 81 \\ 140 \\ 35 \\ 102 \\ 78 \end{bmatrix}$$

$$V = X_{r_1} + F(X_{r_2} - X_{r_3})$$

DE: first solution

- Step 6: Determine trial vector

j	Target Vector	Donor Vector	r	$(r \leq p_c)$	$\delta \neq j$	$\delta = j$	Trial Vector
1	4	4.55	0.3	✓	—	—	4.55
2	0	-0.7	0.9	✗	✓	✗	0
3	1	10.8	0.2	✓	—	—	10.8
4	8	10.7	0.6	✓	—	—	10.7

$$p_c = 0.8, F = 0.85, \delta = 1$$

$$P = \begin{bmatrix} 4 & 0 & 1 & 8 \\ 3 & 1 & 9 & 7 \\ 0 & 3 & 1 & 5 \\ 2 & 1 & 4 & 9 \\ 1 & 2 & 8 & 3 \end{bmatrix} \quad f = \begin{bmatrix} 81 \\ 140 \\ 35 \\ 102 \\ 78 \end{bmatrix}$$

$$u^j = \begin{cases} v^j & \text{if } r \leq p_c \text{ OR } j = \delta \\ x^j & \text{if } r > p_c \text{ AND } j \neq \delta \end{cases}$$

- Step 7: Check bounds, bound if violates

$$U_1 = [4.55 \ 0 \ 10.8 \ 10.7] \rightarrow U_1 = [4.55 \ 0 \ 10 \ 10]$$

$$0 \leq x_i \leq 10$$

$$x = lb \quad \text{if } x < lb$$

$$x = ub \quad \text{if } x > ub$$

First iteration

Value of r_1, r_2, r_3, r, δ for first iteration

$$V = X_{r_1} + F(X_{r_2} - X_{r_3})$$

$$p_c = 0.8, F = 0.85$$

i	Target (P)	r_1	r_2	r_3	Donor (V)	r	δ	Trial (U)
1	[4 0 1 8]	4	2	3	[4.55 -0.7 10.8 10.7]	[0.3 0.9 0.2 0.6]	1	[4.55 0 10 10]
2	[3 1 9 7]	5	1	3	[4.4 -0.55 8 5.55]	[0.3 0.2 0.6 0.4]	4	[4.4 0 8 5.55]
3	[0 3 1 5]	4	2	1	[1.15 1.85 10.8 8.15]	[0.2 0.5 0.4 0.3]	4	[1.15 1.85 10 8.15]
4	[2 1 4 9]	5	3	2	[-1.55 3.7 1.2 1.3]	[0.8 0.3 0.6 0.2]	1	[0 3.7 1.2 1.3]
5	[1 2 8 3]	2	4	1	[1.3 1.85 11.55 7.85]	[0.7 0.5 0.9 0.2]	3	[1.3 1.85 8 7.85]

- Step 8: Evaluate the fitness of bounded trial vectors

$$U = \begin{bmatrix} 4.55 & 0 & 10 & 10 \\ 4.4 & 0 & 8 & 5.55 \\ 1.15 & 1.85 & 10 & 8.15 \\ 0 & 3.7 & 1.2 & 1.3 \\ 1.3 & 1.85 & 8 & 7.85 \end{bmatrix}$$

$$f_U = \begin{bmatrix} 220.70 \\ 114.16 \\ 171.17 \\ 16.82 \\ 130.73 \end{bmatrix}$$

$$u^j = \begin{cases} v^j & \text{if } r \leq p_c \text{ OR } j = \delta \\ x^j & \text{if } r > p_c \text{ AND } j \neq \delta \end{cases}$$

$$f(x) = x_1^2 + x_2^2 + x_3^2 + x_4^2$$

DE: Greedy selection of first iteration

- Step 9: After each iteration, perform greedy selection and update population

$$P = \begin{bmatrix} 4 & 0 & 1 & 8 \\ 3 & 1 & 9 & 7 \\ 0 & 3 & 1 & 5 \\ 2 & 1 & 4 & 9 \\ 1 & 2 & 8 & 3 \end{bmatrix} \quad f = \begin{bmatrix} 81 \\ 140 \\ 35 \\ 102 \\ 78 \end{bmatrix} \quad U = \begin{bmatrix} 4.55 & 0 & 10 & 10 \\ 4.4 & 0 & 8 & 5.55 \\ 1.15 & 1.85 & 10 & 8.15 \\ 0 & 3.7 & 1.2 & 1.3 \\ 1.3 & 1.85 & 8 & 7.85 \end{bmatrix} \quad f_U = \begin{bmatrix} 220.70 \\ 114.16 \\ 171.17 \\ 16.82 \\ 130.73 \end{bmatrix}$$

Population for next iteration

$$P = \begin{bmatrix} 4 & 0 & 1 & 8 \\ 4.4 & 0 & 8 & 5.55 \\ 0 & 3 & 1 & 5 \\ 0 & 3.7 & 1.2 & 1.3 \\ 1 & 2 & 8 & 3 \end{bmatrix} \quad f = \begin{bmatrix} 81 \\ 114.16 \\ 35 \\ 16.82 \\ 78 \end{bmatrix}$$

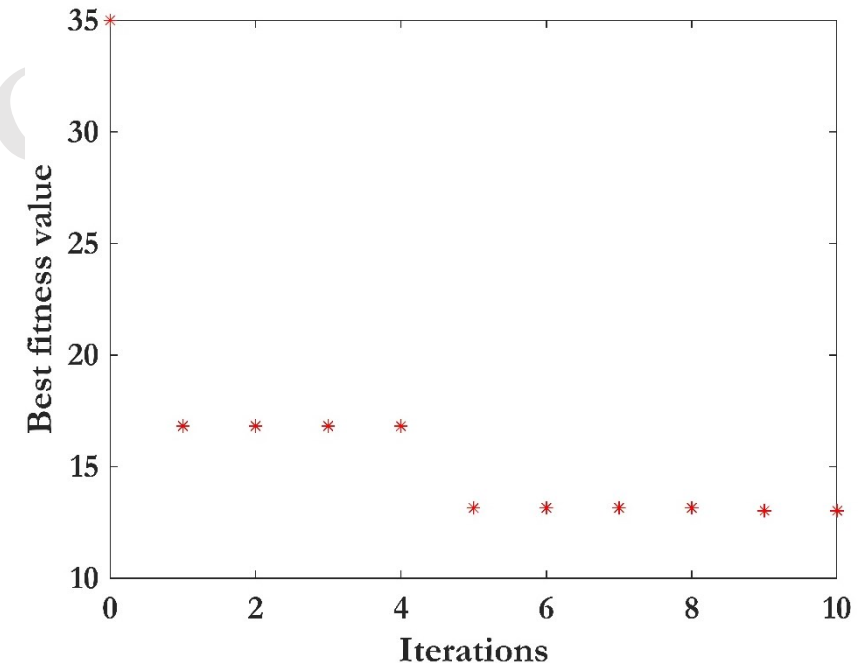
DE: Satisfaction of termination criterion

$$\min f(x) = \sum_{i=1}^4 x_i^2; \quad 0 \leq x_i \leq 10, \quad i = 1, 2, 3, 4$$

After the completion of 10 iterations

$$P = \begin{bmatrix} 0 & 4.18 & 1.2 & 0 \\ 0 & 3.51 & 1.2 & 0 \\ 0 & 3 & 1.2 & 1.65 \\ 0 & 3.10 & 1.2 & 1.40 \\ 0 & 3.68 & 1.2 & 0 \end{bmatrix} \quad f = \begin{bmatrix} 18.89 \\ 13.73 \\ 13.15 \\ 13.02 \\ 15.01 \end{bmatrix}$$

The minimum value of the function is **0**



Comparison between TLBO, PSO and DE

	TLBO	PSO	DE
Phases	Teacher, Learner	No phases (Position and velocity update)	No phases (Mutation and crossover)
Convergence	Monotonic	Monotonic (with g_{best} & p_{best})	Monotonic
Parameters	Population size, termination criteria	Population size, termination criteria, w , c_1 and c_2	Population size, termination criteria, P_c , F
Generation of new solution	Using other solutions, mean and best solution (part of population)	Using velocity vector, p_{best} and g_{best} (need not be the part of population)	Using other solutions (best solution is part of population)
Solution update in an iteration	Twice	Once	Once
Selection	Greedy	Always accept new solution into the population (μ , λ)	Greedy
#FE	$N_p + 2N_p T$	$N_p + N_p T$	$N_p + N_p T$

Further reading

- Differential Evolution – A Simple and Efficient Heuristic for global Optimization over Continuous Spaces, **Journal of Global Optimization**, 11, 341, 1997
- JADE: Adaptive Differential Evolution With Optional External Archive, **IEEE Transactions on Evolutionary Computation**, 13(5), 945-958, 2009
- Differential Evolution Algorithm With Strategy Adaptation for Global Numerical Optimization, **IEEE Transactions on Evolutionary Computation**, 13(2), 398-417, 2009
- Recent advances in differential evolution – An updated survey, **Swarm and Evolutionary Computation**, 27, 1-30, 2016
- Differential evolution with multi-population based ensemble of mutation strategies, **Information Sciences**, 329, 329-345, 2016
- Review of Differential Evolution population size, **Swarm and Evolutionary Computation**, 32, 1-24, 2017

Thank You !!!